

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 16 (Ch. 15)

Financial Markets and Expectations: the
bond market

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
 - Elements of national accounting
2. The short run
 1. Goods markets
 2. Financial markets
 1. Without commercial banks
 2. With commercial banks
 3. IS-LM model
3. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model
 1. Medium and short run equilibrium
 2. Demand and supply shock

Extensions

4. Expectations, financial markets, and economic policies
 1. Financial markets and expectations
 2. Consumption, investment, and expectations
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies

EXPECTATIONS



- Economic policies impact directly the economy, but also **affect expectations**
- Agents' expectations about the future determine **present** choices

THE ROLE OF EXPECTATIONS

Italy Stock Market Index (FTSE Mib)



<https://tradingeconomics.com/italy/stock-market>

Fragile confidence hinders global economic recovery

Political and economic uncertainties drag on business and household mood, Brookings-FT Tiger index shows



Uncertainty about the US presidential election between Kamala Harris and Donald Trump is adding to business and households' concerns © Charles Rex Arbogast/AP

Sam Fleming in London OCTOBER 20 2024

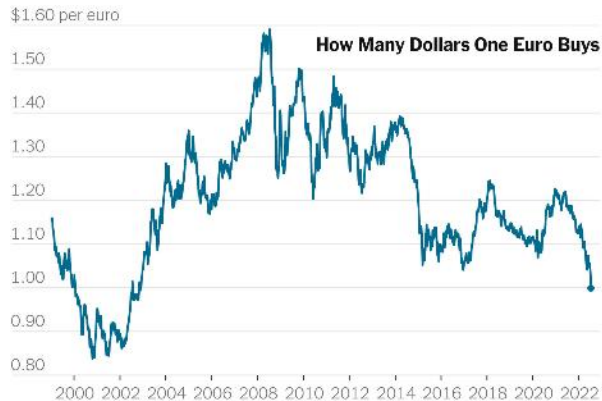
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<https://www.ft.com/content/ba7739b1-af6d-44ff-b202-0182b2134ec4>

The Government reduces taxes on labor

- Permanent vs. Temporary Reduction
- Are you increasing your consumption?

THE OPEN ECONOMY

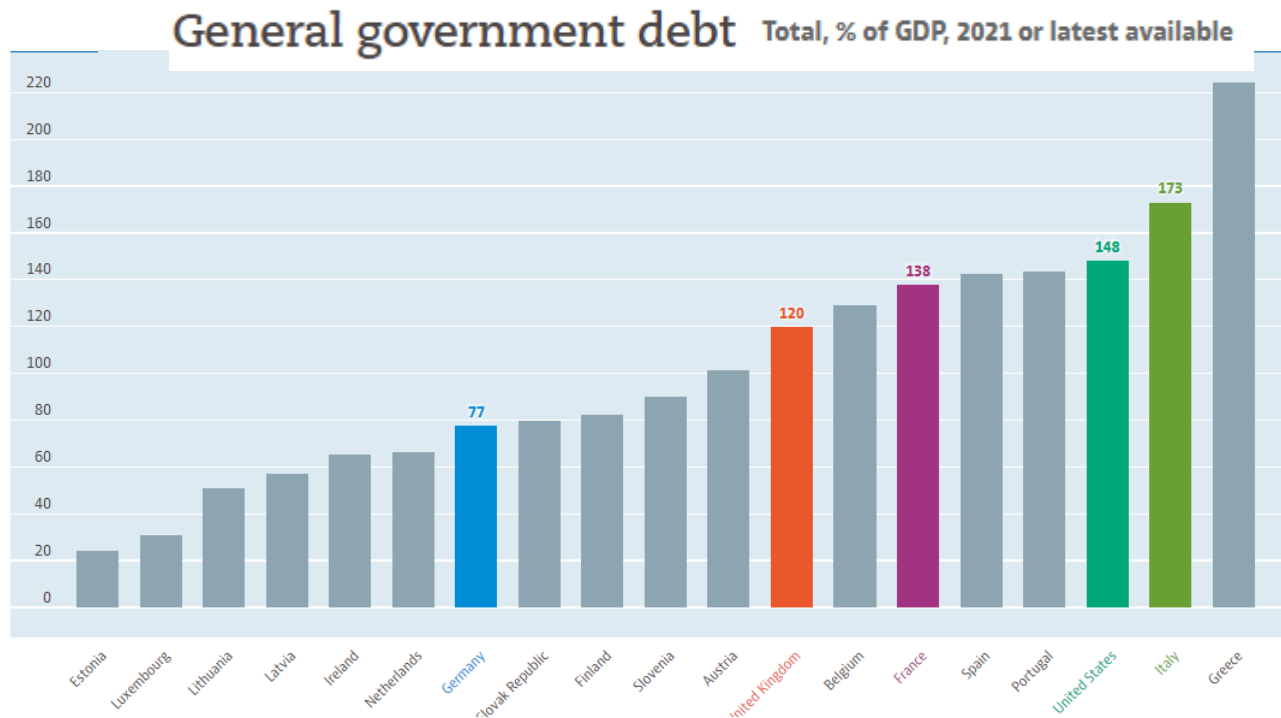


- What happens when countries interact among them, trading goods, services, financial assets?
- What are the implications for monetary and fiscal policies?

SOVEREIGN DEBT



- Is public debt an issue? Is it sustainable?



EXPECTATIONS AND FINANCIAL MARKETS

FINANCIAL MARKETS

Investors trade **financial assets** using their savings

Bonds

Stocks

More complex assets



Financial contracts: a subject lends money **today** in exchange for the promise of receiving additional money in the **future**



Time is a crucial dimension, and agents form expectations about what will happen in the future

Economic conditions

Future economic policies
(FP, MP, structural)

3 main features:

- 1) Liquidity
- 2) Risk level
- 3) Maturity



They influence
financial assets'
prices and returns



FINANCIAL ASSETS

BONDS

- If I buy a bond, as an investor I'm making a **loan** to a borrower
- Today, I pay $\text{€}P_t$, and in the future, after t years, I **will receive the face value** of 100 if the issuer does not default

Certain

STOCKS

- A stock (also called equity) represents **ownership** in a company. If I buy a stock, I own a share of that company and become a shareholder.
- Today, I pay $\text{€}Q_t$, I become shareholders and in the future I **could receive dividends** (a share of the profits)

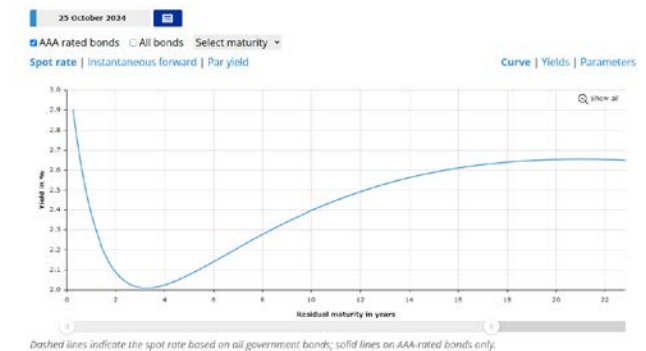
Uncertain

↓
≠ risk level of the two investments

Bund Yields

NAME	COUPON	PRICE	YIELD	1 DAY	1 MONTH	1 YEAR	TIME (EDT)
GTDEM2Y:GOV Germany Bund 2 Year Yield	3.10	100.16	3.01%	-2	-18	+110	10:14 AM
GTDEM5Y:GOV Germany Bund 5 Year Yield	2.40	98.74	2.66%	-1	-10	+69	10:14 AM
GTDEM10Y:GOV Germany Bund 10 Year Yield	2.60	98.11	2.82%	-1	-1	+73	10:14 AM
GTDEM30Y:GOV Germany Bund 30 Year Yield	1.80	74.83	3.10%	-1	+8	+104	10:14 AM

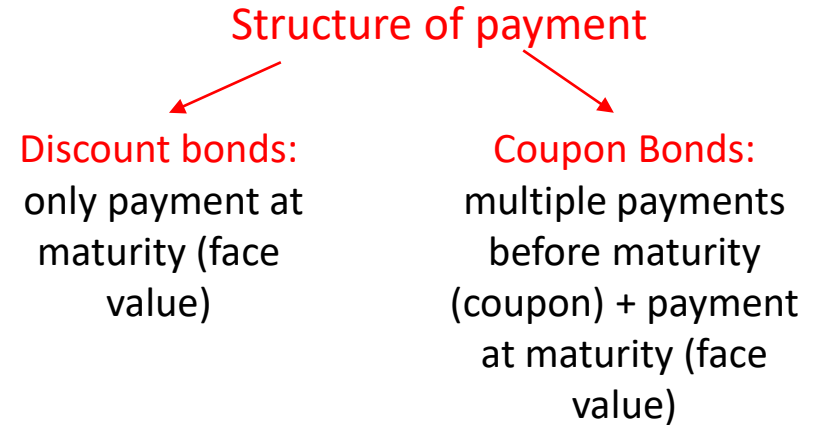
<https://www.bloomberg.com/markets/rates-bonds/government-bonds/germany>



BOND PRICES AND BOND YIELDS

In the real world we have many bonds different in terms of:

1. **Issuer** (public vs. private): Corporate bond vs. Government bond
2. **Maturity**: Length of time over which the bond promises to make payments to the holder



3. Risk

- **Default risk:** Risk that the issuer does not pay the full amount promised
Risk premium based on rating Junk bonds: bonds with high risk of default
- **Price risk:** Uncertainty about the price if you resell the bond BEFORE maturity

Standard & Poor's
AAA(+)
A(+)
BBB(+)
BB(+)
B(+)
CCC
CC

Based on the length of the maturity period (how long until the bondholder is repaid the principal amount) we have:

1. short-term bonds: $m \leq 1$ year

Italian 6 months BOT : bond that promises to make one payment of 100€ in 6 months, it has a maturity of six months

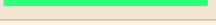
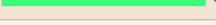
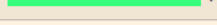
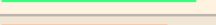
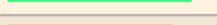
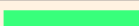
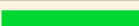
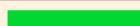
2. long-term bond: $m > 1$ year

Italian 20 years BTP: bond that promises to make one payment of 100€ in 10 years, it has a maturity of 20 years

Feature	Short-Term Bonds	Long-Term Bonds
Maturity	Less than 1 year	More than 1 year (up to 30)
Risk Level	Lower risk	Higher risk
Interest Rate	Lower	Higher
Sensitivity to Inflation	Low	High
Best For	Liquidity & stability	Long-term growth & income

Government Bonds' Ratings

Grade	Moody's	S&P	Fitch
Prime	Aaa	AAA	AAA
High grade	Aa1	AA+	AA+
	Aa2	AA	AA
	Aa3	AA-	AA-
Upper medium grade	A1	A+	A+
	A2	A	A
	A3	A-	A-
Lower medium grade	Baa1	BBB+	BBB+
	Baa2	BBB	BBB
	Baa3	BBB-	BBB-
Non-investment grade speculative	Ba1	BB+	BB+
	Ba2	BB	BB
	Ba3	BB-	BB-
Highly speculative	B1	B+	B+
	B2	B	B
	B3	B-	B-
Substantial risks	Caa1	CCC+	CCC+
	Caa2	CCC	CCC
	Caa3	CCC-	CCC-
Extremely speculative	Ca	CC	CC
			C

Sovereign Ratings List				
	Moody's ratings [+]		S&P ratings [+]	
				Fitch ratings [+]
United States [+]	 Aaa	 AA+	 AAA	
United Kingdom [+]	 Aa3	 AA	 AA-	
Germany [+]	 Aaa	 AAA	 AAA	
France [+]	 Aa2	 AA	 AA	
Japan [+]	 A1	 A+	 A	
Spain [+]	 Baa1	 A	 A-	
Italy [+]	 Baa3	 BBB	 BBB	
Portugal [+]	 Baa2	 BBB	 BBB	
Greece [+]	 Ba3	 BB	 BB	
Ireland [+]	 A2	 AA-	 AA-	

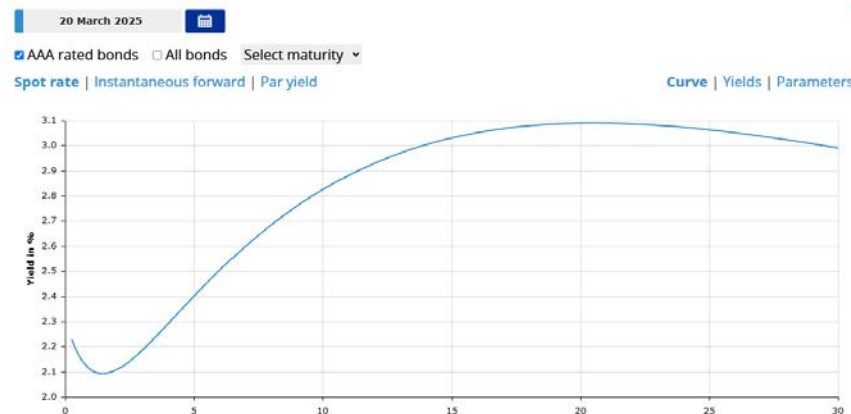
<https://countryeconomy.com/ratings>

- Maturity and risk matter in the determination of the prices of bonds and their interest rate
- Different maturity $\Rightarrow$ different prices $\Rightarrow$ different returns = Yield to maturity (yield)

Yield curve $\leftarrow$
(term structure of interest rates):
on any given day, the
relationship between maturity
and yield

Short-term interest rates:
yields on bonds with a
short maturity (1 year or
less)

Long-term interest rates:
yields on bonds with a
long maturity (more than
1 year)





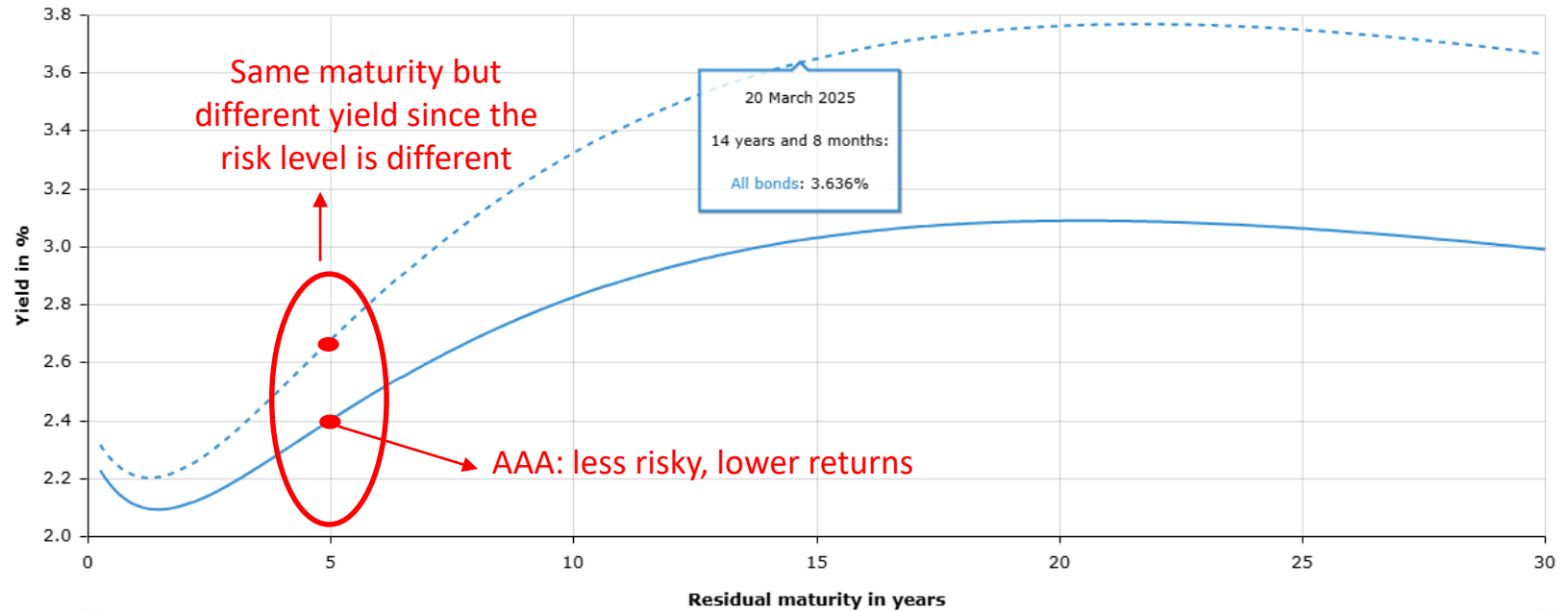
20 March 2025



☒ AAA rated bonds ☒ All bonds Select maturity ▾

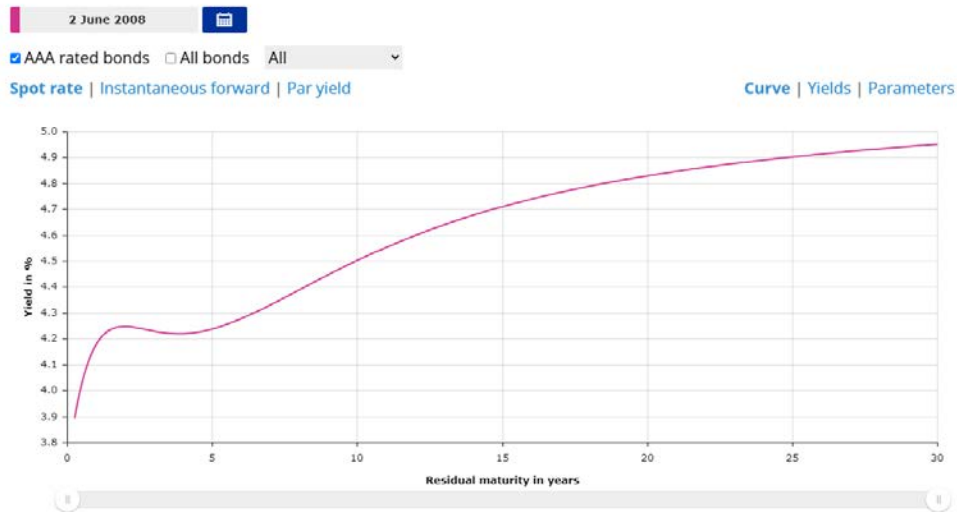
[Spot rate](#) | [Instantaneous forward](#) | [Par yield](#)

[Curve](#) | [Yields](#) | [Parameters](#)

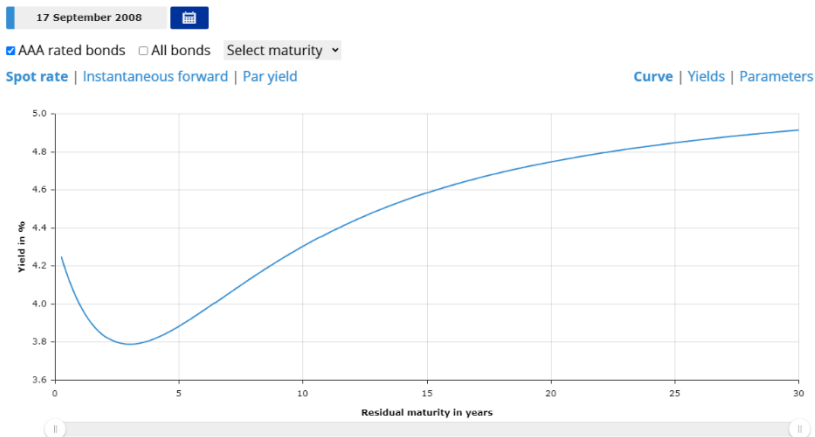


The yield curve changes shape/slope over time

Before the 2008 crisis



2008 financial crisis hits



MP interventions



How is the return of a bond determined?

Assumptions:

- No risk
- Possible to invest in 2 risk free discount bonds (zero coupon) with different maturity and the same face value 100€

1-year bond:
maturity = 1 year from now

2-year bond:
maturity = 2 years from now

- Bought by investors in the primary market of bonds at

$\text{€}P_{1t} \equiv$ price of bond with maturity 1 year
 $\text{€}P_{2t} \equiv$ price of bond with maturity 2 years

→ Determined by D and S of bonds: If investors strongly demand U.S. Treasury bonds, their price increases (and the government can issue them at **lower interest rates** because investors are willing to accept less return for the security they provide)

Today's bond prices reflect the willingness of investors to pay: how much the bond is worth to them => present discounted value of the money they will receive in the future

- 1) **One-year bond:** Pays 100€ in 1 year from now
=> Today, I am willing to pay for it based on how much €100 next year is worth today

$$\text{€}P_{1t} = \frac{100}{1 + i_{1,t}} \text{ €}$$

present discounted value of the payment at maturity: today value of 100€ in one year

CURRENT ONE YEAR INTEREST RATE = CURRENT
SHORT TERM INTEREST RATE

$$\text{€}P_{1t} = f(i_{1,t})$$

Negative relationship btw prices and short term interest rate:

If $\text{€}P_{1t} \uparrow \Rightarrow i_{1,t} \downarrow$

2) **Two-year bond:** Pays 100€ in 2 years from now

=> Today, I am willing to pay for it based on how much €100 in two years is worth today

Today price can be obtained from the

Arbitrage condition btw financial assets:

In order to have a market for all existing financial assets, the expected returns on two assets must be equal

Assumption confirmed by the empirical observation: both types of bonds are traded at any given time => offer the same expected one-year return

HP: you have 1€ to invest



Today: t
Investment choice

1 year from now: $t + 1$
Return for each € invested at t

n. bought bonds

Bond 1:
1-year bond

$$\frac{1}{\epsilon P_{1t}}$$



$$\frac{100\epsilon}{\epsilon P_{1t}} = 1 + i_{1,t} \epsilon \rightarrow \text{Certain return from the short term investment}$$

Bond 2:
2-year bond

$$\frac{1}{\epsilon P_{2t}}$$



$$\frac{1}{\epsilon P_{2t}} \epsilon P_{1t+1}^e \epsilon \rightarrow \text{Expected return from the long term investment}$$

Next year the bond will have one more year before maturity => it will be a one-year bond and you can sell it at a price ϵP_{1t+1}^e = Expected price of a one-year bond in $t + 1$

To have equilibrium in the bonds market with bonds with different maturity traded at any given time the expected one-year return of alternative bonds must be the same

$$(1 + i_{1t}) = \frac{\epsilon P_{1t+1}^e}{\epsilon P_{2t}} \equiv \text{arbitrage condition btw one and two year bonds}$$

By rearranging we obtain today price of a two – year bond

$$\epsilon P_{2t} = \frac{\epsilon P_{1t+1}^e}{1 + i_{1t}} = \frac{1}{1 + i_{1t}} \frac{\epsilon 100}{1 + i_{1,t+1}^e}$$

PDV of the expected
price next year

The price of a one-year bond in t+1 will be PDV of €100 received in two years
=> The face value must be discounted using the expected future short-term interest rate = $i_{1,t+1}^e$

$$\epsilon P_{2t} = \frac{\epsilon 100}{(1 + i_{1,t})(1 + i_{1,t+1}^e)}$$

present discounted value of the payment at maturity:
today value of 100€ in two years
=> the face value must be discounted using the current short-term interest rate ($i_{1,t}$) and the expected future short-term interest rate ($i_{1,t+1}^e$)

What happens if the arbitrage condition does not hold?

If $(1 + i_{1t}) < \frac{\epsilon P_{1t+1}^e}{\epsilon P_{2t}}$ $\Rightarrow$ more convenient for investors to buy 2 – year bonds $\Rightarrow \epsilon P_{2t} \uparrow$
 $\Rightarrow$ Expected return on 2–year bond $\downarrow$

If $(1 + i_{1t}) > \frac{\epsilon P_{1t+1}^e}{\epsilon P_{2t}}$ $\Rightarrow$ more convenient for investors to buy 1– year bonds $\Rightarrow \epsilon P_{2t} \downarrow$
 $\Rightarrow$ Expected return on 2–year bond $\uparrow$

- In general, the price of a bond today is the present value of the payment promised at maturity.
=> There is a negative relationship between bond prices and short-term interest rates (both current and expected).

- 1-year bond: $\text{€}P_{1t} = f(i_{1,t})$

- 2-year bond: $\text{€}P_{2t} = f(i_{1,t}, i_{1,t+1}^e)$

...

- *m-year bonds*: $\text{€}P_{mt} = f(i_{1,t}, i_{1,t+1}^e, i_{1,t+2}^e, i_{1,t+3}^e, \dots)$

current short-term
interest rate

All the expected future short-term interest
rates from issuance to repayment

True or false?

The price of a two-year bond today is higher if investors expect current short-term interest rates to be lower than future short-term interest rates.

False

What is the price of a 2- year bond if for example:

- $i_t = i_{t+1}^e = 5\% \Rightarrow$ no changes in MP at t+1

$$\text{€}P_{2t} = \frac{100}{(1+0.05)^2} = 90.7$$

- $i_t = 5\%, i_{t+1}^e = 8\% \Rightarrow$ expected contractionary MP at t+1

$$\text{€}P_{2t} = \frac{100}{(1+0.05)(1+0.08)} = 88.18$$

FROM BOND PRICES TO BOND YIELDS

Based on its maturity, a bond will have a price associated with a corresponding interest rate (yield to maturity)

Yield (to maturity) of an m -year bond (m -year interest rate)

Interest rate associated with a bond with maturity m years
(average annual interest rate paid by the bond)

=

Constant annual interest rate that makes the **price of the bond today**
equal to the **expected present value of future payments promised**



For a zero coupon bond the only payment is the final reimbursement

$$i_{mt} : \text{€}P_{mt} = \frac{\text{€}100}{(1+i_{mt})^m}$$

Yield to maturity for a

- 1-year bond: $i_{1t} : \epsilon P_{1t} = \frac{\epsilon 100}{1+i_{1t}}$

$$1 + i_{1t} = \frac{\epsilon 100}{\epsilon P_{1t}} \Rightarrow i_{1t} = \frac{\epsilon 100 - \epsilon P_{1t}}{\epsilon P_{1t}} \rightarrow \text{Current short term interest rate (1 year interest rate): } \% \Delta \text{ btw paid price and reimbursement}$$

- 2-year bond: $i_{2t} : \epsilon P_{2t} = \frac{\epsilon 100}{(1+i_{2t})^2}$

$$(1 + i_{2t})^2 = \frac{\epsilon 100}{\epsilon P_{2t}} \Rightarrow i_{2t} = \sqrt{\frac{\epsilon 100}{\epsilon P_{2t}}} - 1$$

- 3-year bond: $i_{3t} : \epsilon P_{3t} = \frac{\epsilon 100}{(1+i_{3t})^3}$

$$(1 + i_{3t})^3 = \frac{\epsilon 100}{\epsilon P_{3t}} \Rightarrow i_{3t} = \sqrt[3]{\frac{\epsilon 100}{\epsilon P_{3t}}} - 1$$

Long term interest rates

....

QUICK CHECK

If an investor pays €88.18 today for a 2-year bond and holds the asset in their portfolio until maturity, what is the approximate annual return?

- a) 5%
- b) 6.5%**
- c) 7.2%
- d) 4.8%

Having a 2-year bond the yield (to maturity) or 2-year interest rate i_{2t} is the constant interest rate that satisfy the following equation

$$\text{€}P_{2t} = \frac{\text{€}100}{(1 + i_{2t})^2}$$

$$\text{If } \text{€}P_{2t} = 88.18\text{€}$$

$$\begin{aligned}\Rightarrow (1 + i_{2t})^2 &= \frac{100}{88.18} \\ \Rightarrow (1 + i_{2t}) &= \sqrt{1.134} \\ \Rightarrow i_{2t} &\approx 6.5\%\end{aligned}$$

By holding the asset in the portfolio for two years, we obtain an average yearly return of 6.5%.

What is the relationship btw short and long term interest rates?

In the case of two bonds we know that

1) Bond price definition

$$€P_{2t} = \frac{€100}{(1 + i_{1t})(1 + i_{1t+1}^e)}$$

2) Yield definition

$$€P_{2t} = \frac{€100}{(1 + i_{2t})^2}$$

They must be equal

$$\frac{€100}{(1 + i_{1t})(1 + i_{1t+1}^e)} = \frac{€100}{(1 + i_{2t})^2}$$

$$(1 + i_{2t})^2 = (1 + i_{1t})(1 + i_{1t+1}^e)$$

Exact relationship between long-term and short-term (current and expected) interest rates: if short-term interest rates $\uparrow$, long-term interest rates $\uparrow$ too

$$\text{Ex: } i_t = 5\% \quad i_{t+1}^e = 8\% \Rightarrow (1 + i_{2t})^2 = (1 + 0.05)(1 + 0.08)$$

$$1 + i_{2t} = 1.065 \Rightarrow i_{2t} = 0.065 = 6.5\%$$

The previous relationship can be approximated using a logarithmic transformation by applying logs to both sides

$$\ln(1 + i_{2t})^2 = \ln(1 + i_{1t})(1 + i_{1t+1}^e)$$

$$2 \ln(1 + i_{2t}) = \ln(1 + i_{1t}) + \ln(1 + i_{1t+1}^e)$$

Approximating:

$$2i_{2t} \approx i_{1t} + i_{1t+1}^e$$

$$i_{2t} \approx \frac{1}{2} (i_{1t} + i_{1t+1}^e)$$



Approximated relationship between long-term and short-term interest rates:
the 2 year interest rate is approximately the average between current and future expected one-year interest rates

$$\text{Ex: } i_t = 5\% \quad i_{t+1}^e = 8\% \Rightarrow i_{2t} \approx \frac{5\% + 8\%}{2} = 6.5\%$$

OBSERVATIONS

1. Long-term interest rates reflect both current and expected future short-term interest rates
=> Based on long-term and current short-term interest rates, we can **infer what investors expect** for future short-term interest rates

Given

$$i_{2t} \approx \frac{1}{2}(i_{1t} + i_{1t+1}^e) \Rightarrow i_{1t+1}^e = 2i_{2t} - i_{1t}$$

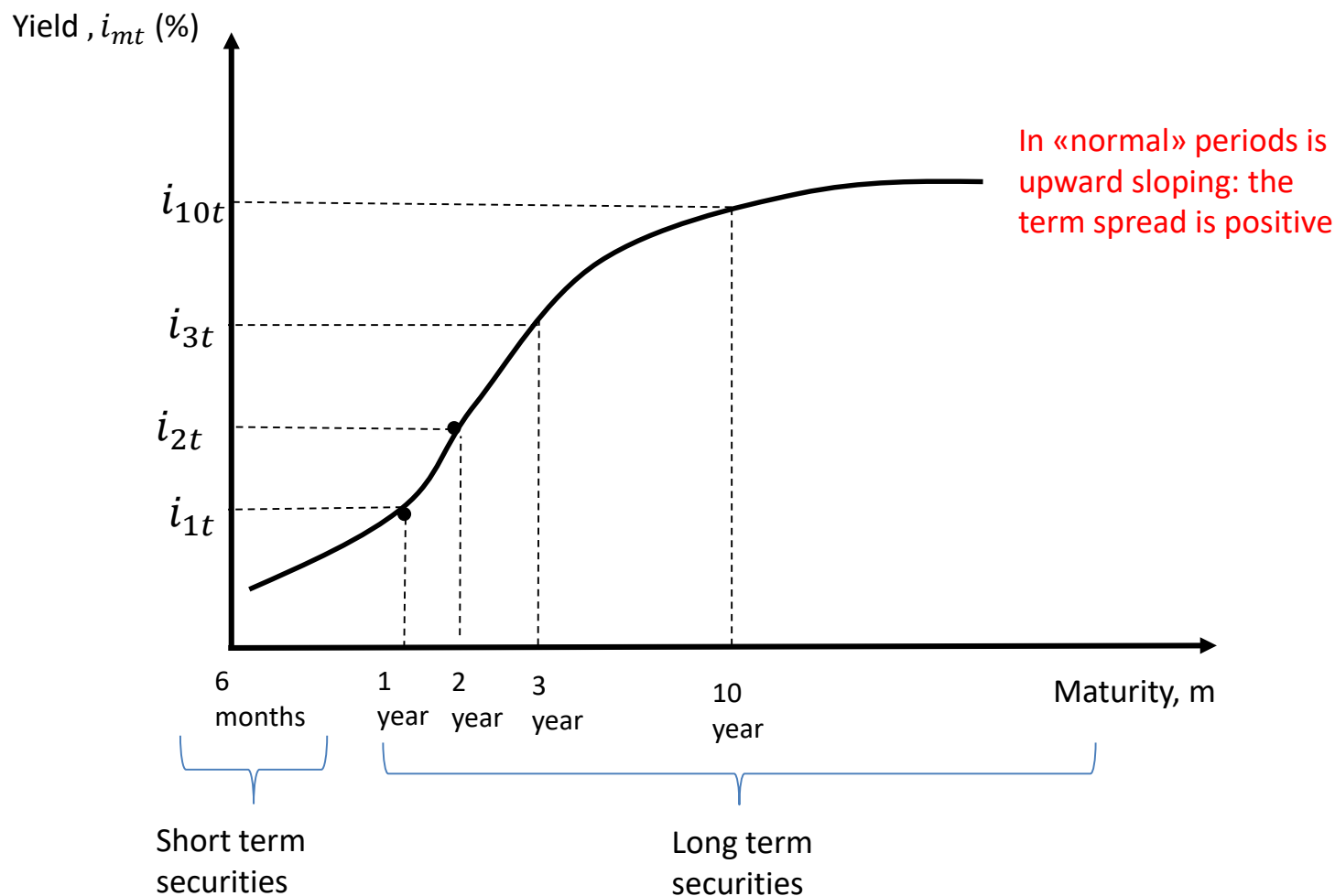


Short term interest rate expected by investors for the next period

2. The relationship can be **generalized for each maturity**

$$i_{mt} \approx \frac{1}{m}(i_{1t} + i_{1t+1}^e + i_{1t+2}^e + \dots + i_{1t+m-1}^e)$$

3. **Yield curve** (term structure of the interest rates) : graphical representation, at a given point in time, of the relationship between maturity (horizontal axis) and yield (vertical axis).



4. Financial operators monitor the yield curve of an economy and analyze its:

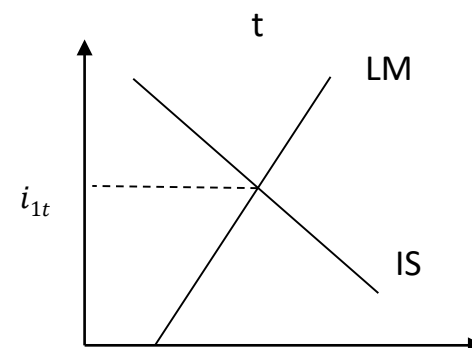
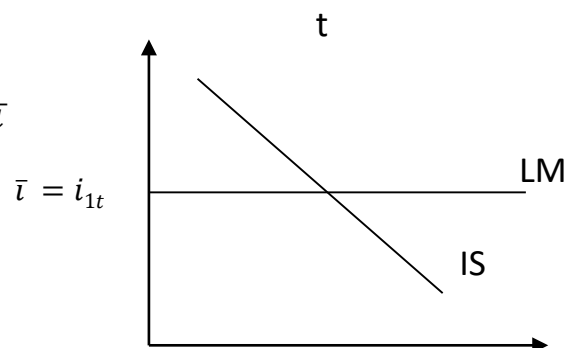
- Position (starting point)
- Slope (sign of the term spread)
- Shifts

1. **Position** depends on the current short term interest rate i_{1t}



Influenced by current MP

CB chooses the policy rate $\bar{i}$



2. **Slope** depends on the term spread: the difference btw expected future short term interest rate and current short term interest rate



Influenced by current and expected future MP

Slope depends on the sign of

$$i_{2t} - i_{1t} =$$

$$\frac{1}{2}(i_{1t} + i_{1t+1}^e) - i_{1t} = \frac{1}{2}i_{1t} + \frac{1}{2}i_{1t+1}^e - i_{1t} =$$

$$\frac{i_{1t+1}^e - i_{1t}}{2}$$

Positive slope

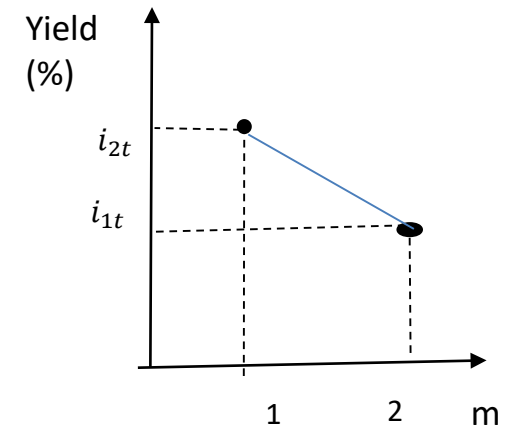
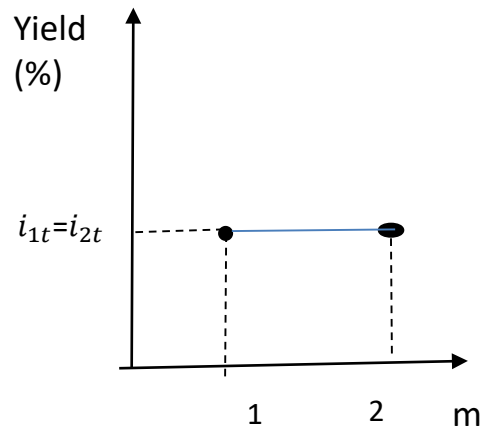
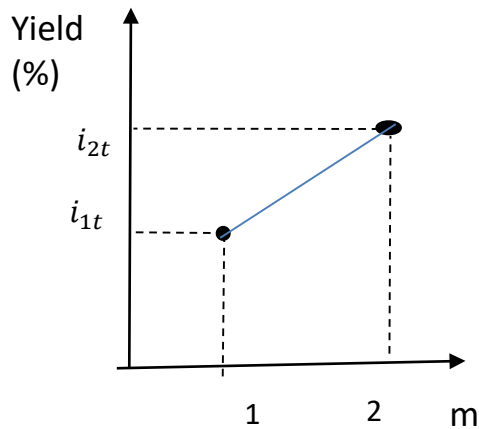
$$i_{2t} > i_{1t} \Leftrightarrow i_{1t+1}^e > i_{1t}$$

Flat

$$i_{2t} = i_{1t} \Leftrightarrow i_{1t+1}^e = i_{1t}$$

Negative slope

$$i_{2t} < i_{1t} \Leftrightarrow i_{1t+1}^e < i_{1t}$$



HP:

- 3 risk free assets: 1 year, 2 year, 3 year bond

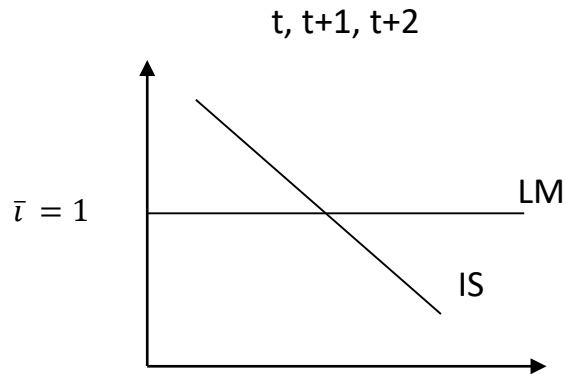
=> three yield to maturity: i_{1t}, i_{2t}, i_{3t}

=> Yield curve with three points

- CB chooses the policy rate $\bar{i}=1\%$

How is the yield curve influenced by expectations?

1. Agents do not expect any intervention by the central bank over the next three years

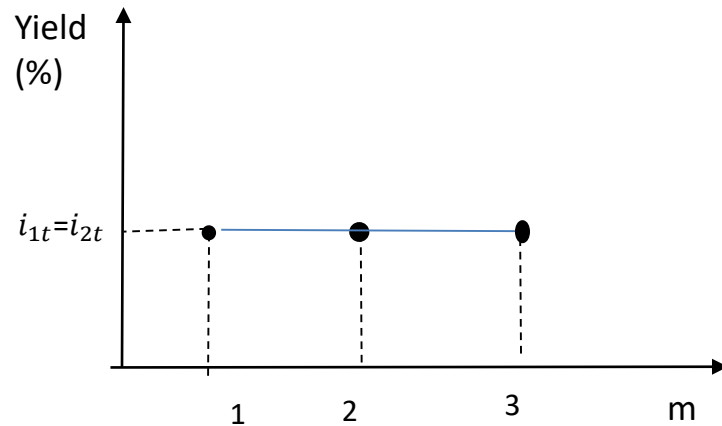


Same policy rate for three years

$$i_{1t} = \bar{i} = 1$$

$$i_{2t} = \frac{1}{2}(i_{1t} + i_{1t+1}^e) = \frac{1}{2}(1 + 1) = 1$$

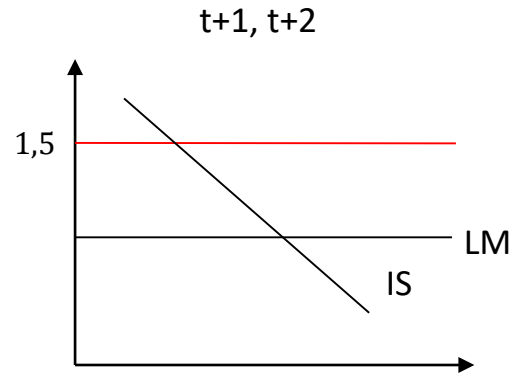
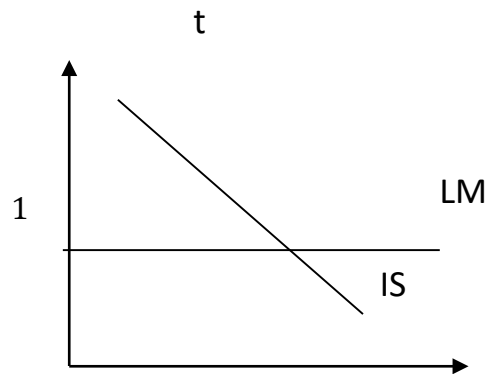
$$i_{3t} = \frac{1}{3}(i_{1t} + i_{1t+1}^e + i_{1t+2}^e) = \frac{1}{3}(1 + 1 + 1) = 1$$



Flat yield curve:

- Short term rates = long term rates
- No term spread
- Current short term rates = expected future short term rates
- No changes in MP

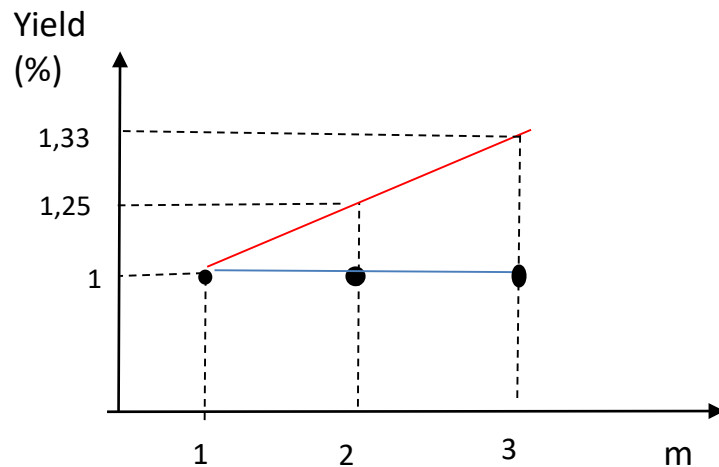
2. Agents expect the central bank will increase the policy rate from the next period at 1,5%



$$i_{1t} = \bar{i} = 1$$

$$i_{2t} = \frac{1}{2}(i_{1t} + i_{1t+1}^e) = \frac{1}{2}(1 + 1,5) = 1,25$$

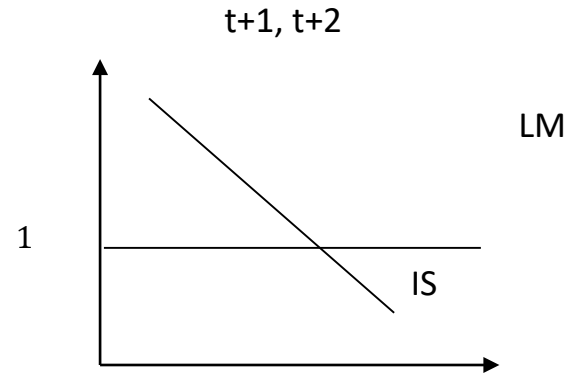
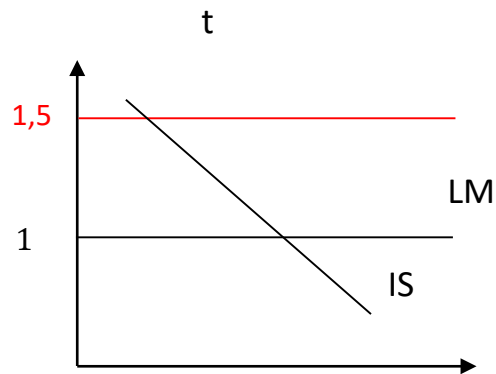
$$i_{3t} = \frac{1}{3}(i_{1t} + i_{1t+1}^e + i_{1t+2}^e) = \frac{1}{3}(1 + 1,5 + 1,5) = 1,33$$



Same starting point but positive slope

- Short term rates < long term rates
- Positive term spread
- Current short term rates < expected future short term rates

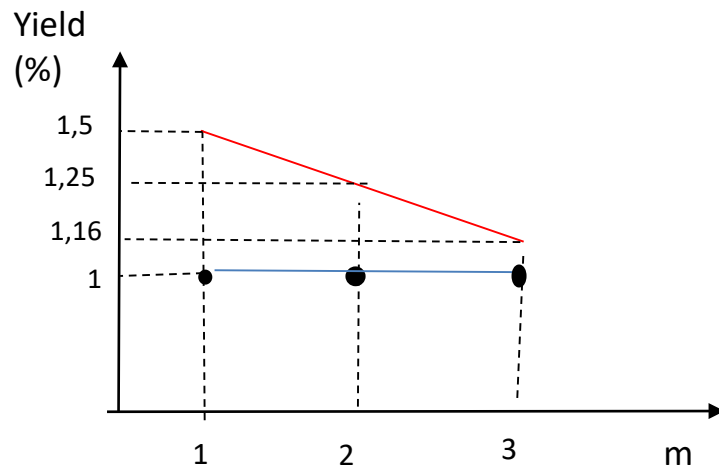
3. The CB increases the current policy rate from 1 to 1,5% only for the current period



$$i_{1t} = \bar{i} = 1,5$$

$$i_{2t} = \frac{1}{2}(i_{1t} + i_{1t+1}^e) = \frac{1}{2}(1,5 + 1) = 1,25$$

$$i_{3t} = \frac{1}{3}(i_{1t} + i_{1t+1}^e + i_{1t+2}^e) = \frac{1}{3}(1,5 + 1,5 + 1) = 1,16$$



Higher starting point and negative slope

- Short term rates > long term rates
- Negative term spread
- Current short term rates > expected future short term rates

ECB yield curve

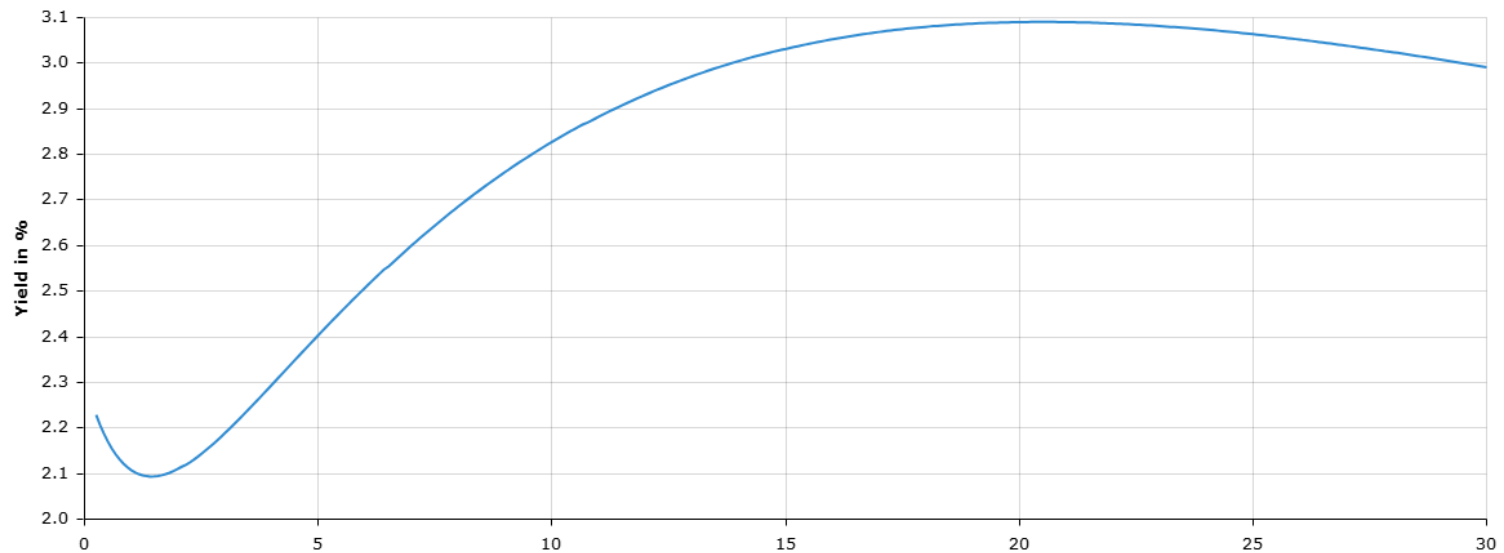
20 March 2025



☒ AAA rated bonds ☐ All bonds Select maturity ▾

Spot rate | Instantaneous forward | Par yield

Curve | Yields | Parameters



https://www.ecb.europa.eu/stats/financial_markets_and_interest_rates/euro_area_yield_curves/html/index.en.html

SUMMARY

- Bond prices for different maturities
- Bonds' yields
- Provides information about people's expectations
- Interpreting the yield curve:
 - Slope
 - Position
- What's next?
 - Risk aversion and the yield curve

RISK AVERSION AND THE YIELD CURVE

In the reality securities are not risk free

We reintroduce risk assuming we have

1-year bond: no risk

We know with certainty the amount received 1 year from now (face value)

=> certain return $(1 + i_{1t})$

2-year bond: risky

We don't know for sure the amount we can get 1 year from now

Uncertainty over the price of the bond 1-year from now

Risk premium: x

1€ to invest

Short term investment

return on a 1-year bond
 $1 + i_{1t}$

Long term investment

Nominal value in one year uncertain:
I do not know the market price of
the bond



Expected one-year return on a 2-year bond

$$\frac{1}{\text{€}P_{2t}} \quad \text{€}P_{1t+1}^e = \underbrace{1 + i_{1t}}_{\text{Return on a 1-year bond}} + \underbrace{x}_{\text{Risk premium}}$$

n. bonds
bought with
1€

Agents must be rewarded for the higher risk
they face

- Today price of a 2-year bond will be:

$$\text{€}P_{2t} = \frac{\text{€}P_{1t+1}^e}{1 + i_{1t} + x}$$

$$\text{€}P_{2t} = \frac{\text{€}100}{(1 + i_{1t} + x)(1 + i_{1t+1}^e)} \longrightarrow \text{Price as expected present discounted value taking into account risk}$$

- Using the definition of yield to maturity we know that

$$\text{€}P_{2t} = \frac{\text{€}100}{(1 + i_{2t})^2}$$

Considering the two conditions together we have:

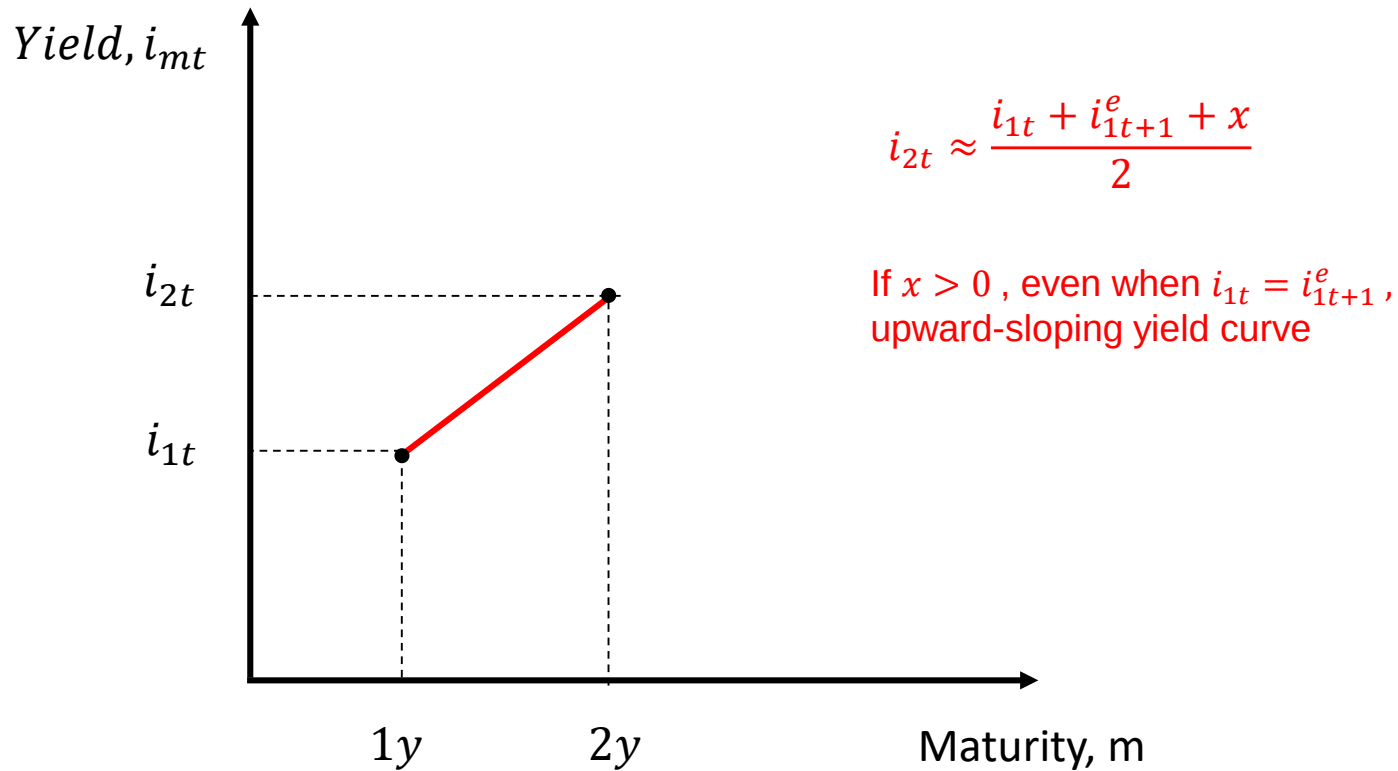
$$\frac{\text{€}100}{(1 + i_{1t} + x)(1 + i_{1t+1}^e)} = \frac{\text{€}100}{(1 + i_{2t})^2}$$

$$(1 + i_{2t})^2 = (1 + i_{1t} + x)(1 + i_{1t+1}^e)$$

$$i_{2t} \approx \frac{1}{2}(i_{1t} + i_{1t+1}^e + x)$$

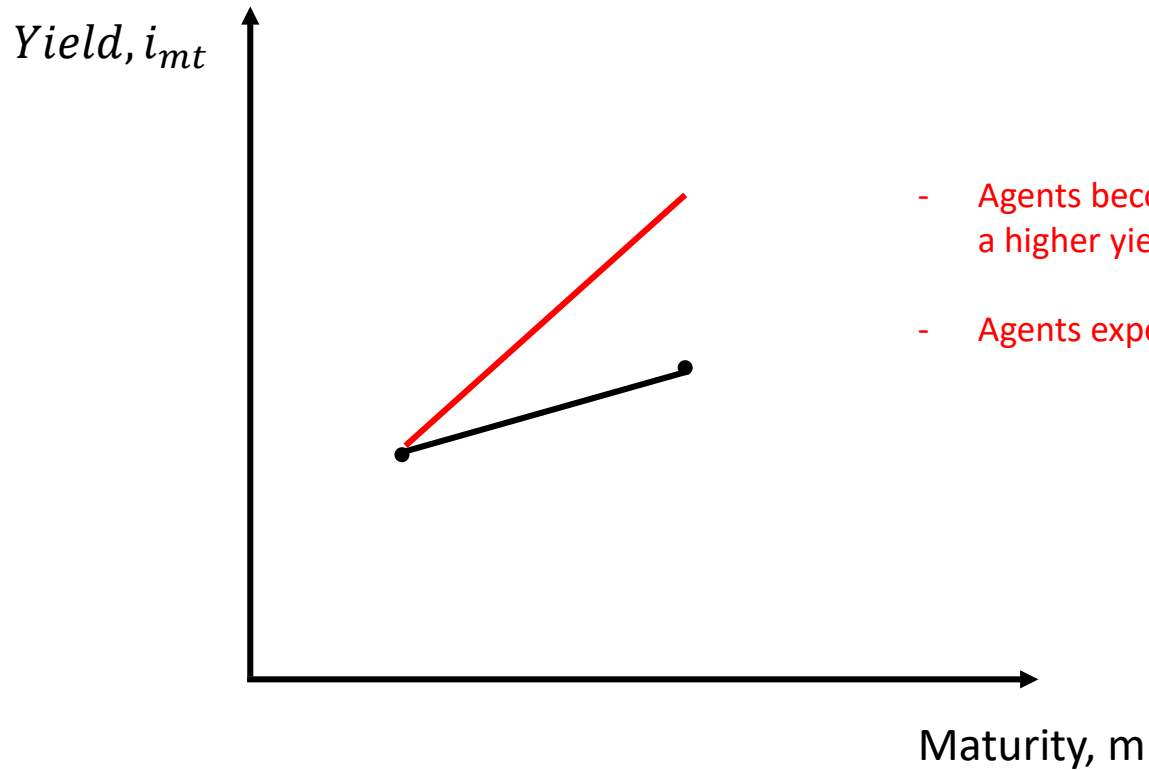
YIELD CURVE WITH RISK

What is the shape of the yield curve if agents do not expect any change in MP?



YIELD CURVE WITH RISK

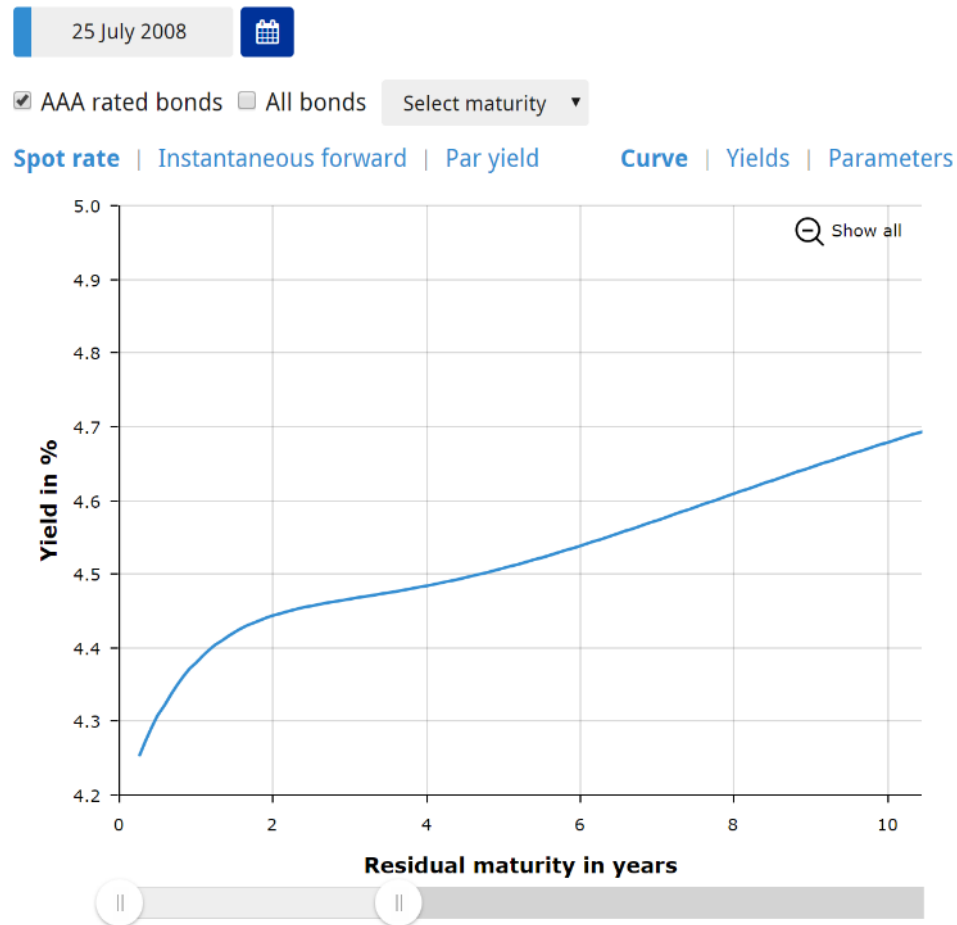
When does the yield curve become steeper?



- Agents become more risk-averse: they demand a higher yield to invest over longer periods
- Agents expect contractionary MP

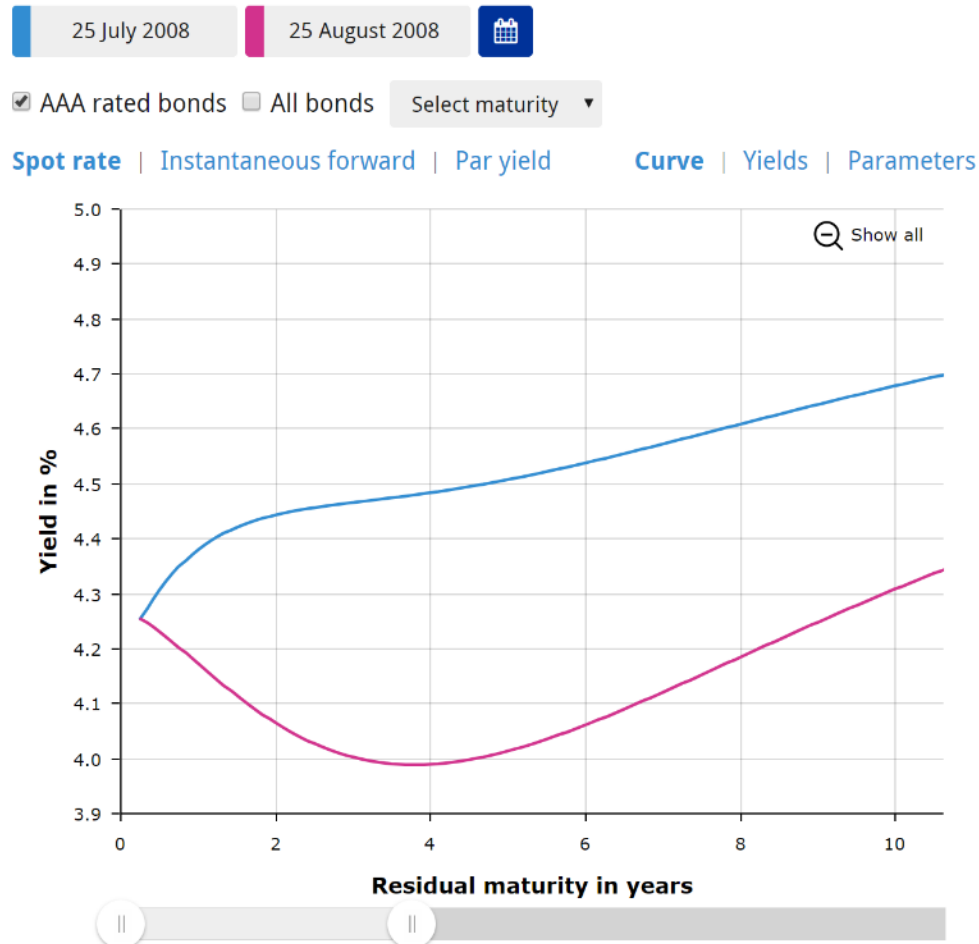
Yield curve

Collapse of Lehman Brothers (Sept 15, 2008)



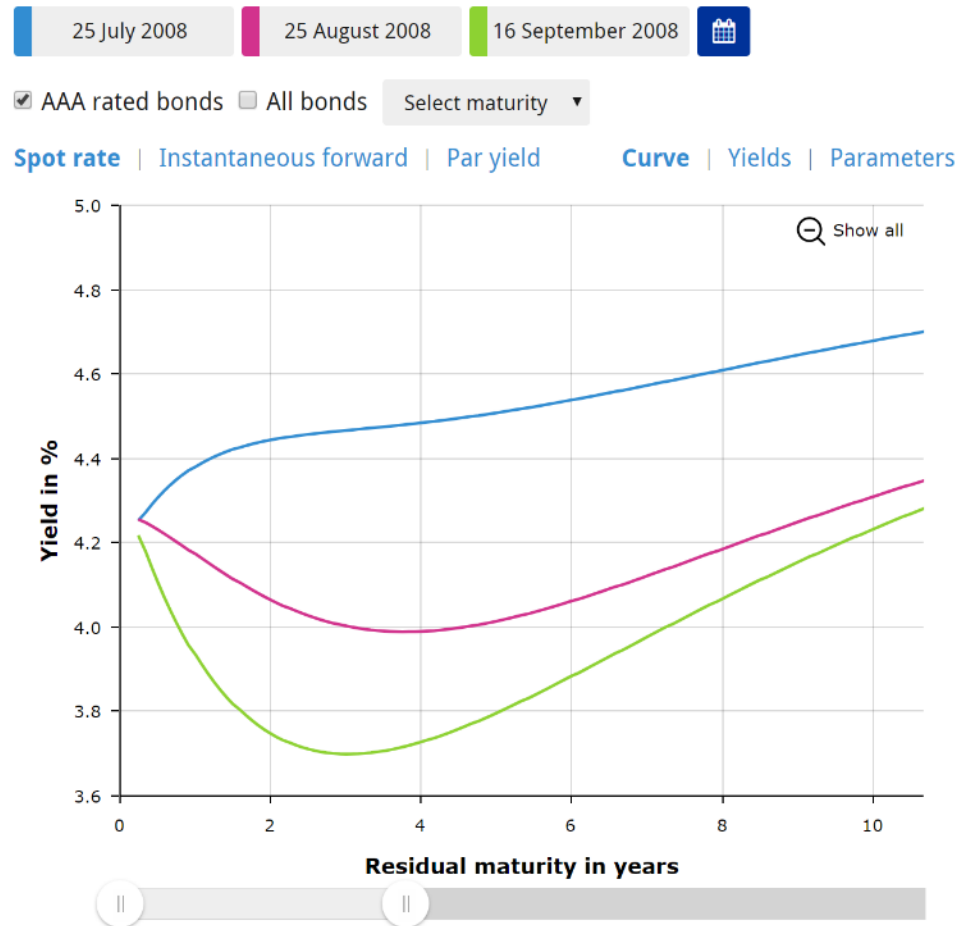
Yield curve

Collapse of Lehman Brothers (Sept 15, 2008)



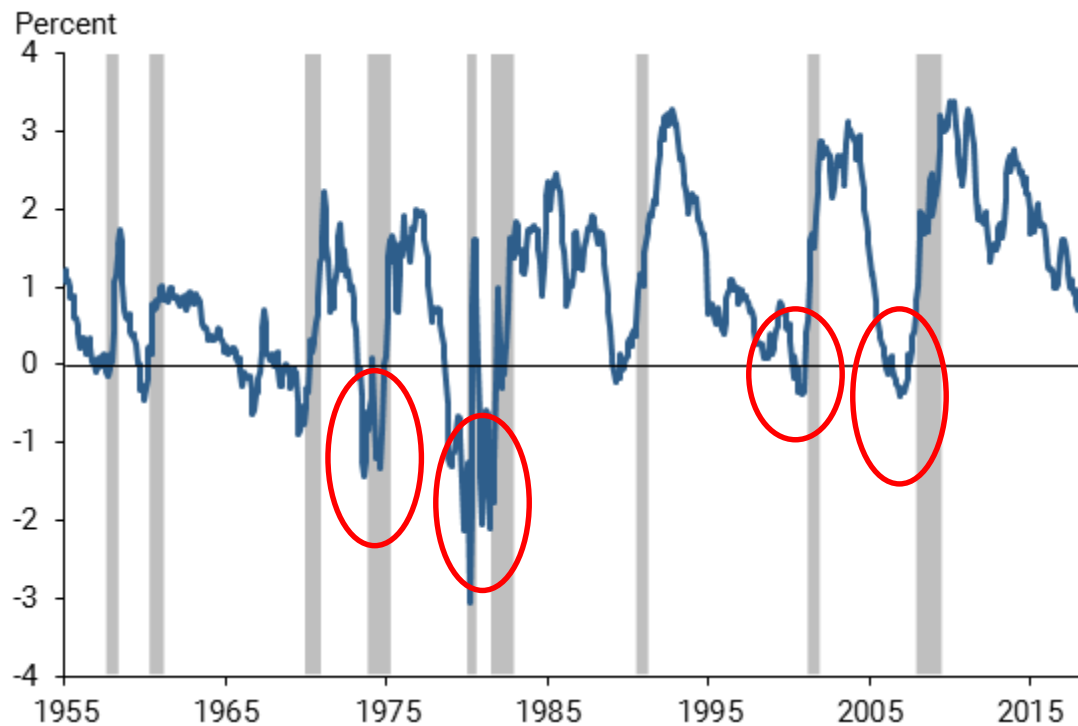
Yield curve

Collapse of Lehman Brothers (Sept 15, 2008)



....WAITING FOR RECESSION

$$\text{term spread} = i_{10t} - i_{1t}$$

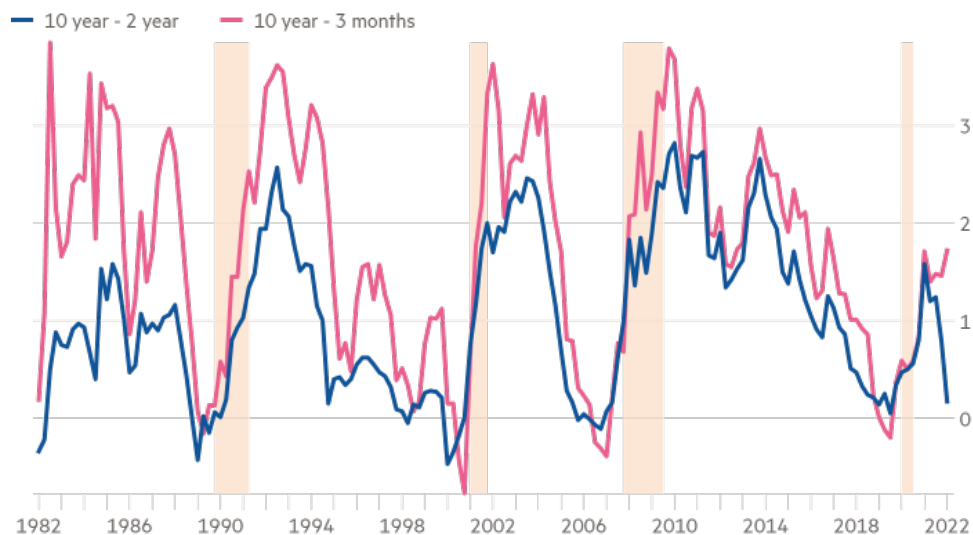


The term spread (the difference between long-term and short-term interest rates) is an accurate predictor of future economic activity

In the last 60 years before every U.S. recession the term spread turned negative and we had an inverted yield curve

Four for four

US Treasury curves, per cent; recessions highlighted



© FT

David Kelly, chief strategist at JPMorgan Asset Management:

*“An inverted yield curve doesn’t do much to the economy, but it’s a very bad sign. The only reason you’d buy a long-term bond at a lower yield than a short-term one was if you thought yields were going to fall . . . **this usually happens when most people think the Fed has gone too far or will go too far.**”*

Robert Armstrong, [“The yield curve might be wrong”](#), Financial Times, March 22, 2022

QUICK CHECK

If individuals are risk averse and the yield curve of a country is upward sloping, we can conclude that:

- A. Agents expect a contractionary monetary policy in the future
- B. Agents expect no change in the monetary policy in the future
- C. **Both a) and b) are compatible with an upward-sloping yield curve**
- D. None of the above

QUICK CHECK

What happens to the yield of a bond of any maturity if its price increases?

- A) The yield increases
- B) The yield stays the same
- C) The yield decreases**
- D) The maturity shortens

QUICK CHECK

An upward-sloping yield curve suggests that:

- A) Investors expect expansionary monetary policy in the future
- B) Short-term bonds have higher yields than long-term bonds
- C) Investors expect inflation in the future**
- D) Bonds are becoming risk-free

QUICK CHECK

If the yield curve is inverted (downward-sloping also only for short maturities), it may signal:

- A) A strong economic recovery
- B) Higher future inflation
- C) An upcoming recession**
- D) That long-term bonds are more risky

TAKE AWAY AND KEY CONCEPTS

- Bonds
- Stocks
- Liquidity
- Maturity
- Risk (default risk, price risk, risk premium)
- Zero-coupon bond
- Face value
- Present discounted value
- Arbitrage condition
- Bond prices for different maturities
- Bonds' yields
 - Short-term interest rate
 - Long-term interest rate
- Yield to maturity
- Yield curve
- Slope (positive/negative/flat)
- Term spread
- Expectations hypothesis
- Normal yield curve
- Inverted yield curve